



USC

CSCI 585 - Database Systems

Units: 4

Fall 2026 - Lecture Monday 5-7:20 pm

Discussion Monday 7:30-8:20 pm

Room: SGM 123

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Office: GCS 505E

Office Hours: Mon 10-12:30 pm, Mon 2-4 pm

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Course Description

The database management industry is a substantial and rapidly growing sector. It is driven by the exponential increase in data, the widespread adoption of cloud computing, and advancements in AI and analytics. In 2023, the global Database Management System (DBMS) market was valued between 60 to 100 billion dollars. While NoSQL databases are growing rapidly, relational databases with ACID transactions still hold a significant share estimated at 60-70%. CSCI 585 introduces students to techniques that enable both SQL and NoSQL DBMSs to scale in the cloud (both within a data center and across data centers), physical organization of data for enhanced performance and availability, cache augmented database systems and memory management, disaggregated database systems, transactional storage managers, and data analytics. The discussion sections introduce basics of transactions with ACID properties, a file system, conceptual and relational data models, SQL and Gremlin, persistent index structures such as hashing and B+-tree, relational algebra, normal forms, fundamentals of a magnetic disk drive and solid state drives including their working details, algorithms such as LRU, concurrency control protocols such as locking and time-stamp, and crash recovery protocols such as logging. An understanding of these concepts is fundamental when using AI to author software for data driven applications.

Guest lecturers from companies such as LinkedIn, eBay, and Databricks will share their insights on building database systems.

- September 14, 2026: Dr. Hieu Nguyen of eBay presents transactional storage manager FoundationDB, its use to implement a graph database, Gremlin query language, and an execution engine process Gremlin queries.
- October 19, 2026: Dr. Haoyu Huang of Databricks presents Lakebase, a new open database architecture that decouples transactional compute from storage while placing data in low-cost cloud object stores in open format.

Learning Objectives

By the end of the semester, students should have a grasp of fundamental concepts, design decisions, protocols and algorithms to build a database system using off-the-shelf software and hardware components. They should understand analytics and ACID semantics and their application to both storage managers, database systems, and applications. Finally, they should be able to provide an abstraction of today's hardware in support of database systems.

Prerequisite(s): Principles of Software Development (CSCI 201).

Recommended Preparation: A good understanding of in-memory data structures such as records and their organizations in arrays, trees, hash tables, etc. Knowledge of relational databases, SQL, relational algebra and physical database design is required. We will cover the relational data model+algebra, SQL, physical database design in the discussion sections.

Course Notes:

- All lecture material will be posted on the USC Brightspace.
- Grading breakdown
 - Exam 1, Oct 12, 2026: 50%
 - Exam 2, Nov 30, 2026: 50%
- Each Exam counts toward 50% of their grade.
- This course has no comprehensive final exam. Exam 2 is held during the last lecture and covers material presented since Exam 1.

Communication

This course uses the USC Brightspace system to disseminate lectures and assigned readings, and make announcements.

Technological Proficiency and Hardware/Software Required

Required Readings and Supplementary Materials

This course assumes students are Internet savvy and capable of using it as a digital library. This includes use of ACM/IEEE/Springer digital libraries. All USC students have access to these digital collections automatically.

Grading Breakdown

Based on prior experience teaching this course, students who attend lectures do very well. In specific, students who skip lectures do poorly on Exam 1 and spend the second half of the semester concerned with their grade. To prevent this, attend lectures and participate by asking questions. *Each Exam is worth 50% of the overall grade.*

Assessment Tool (assignments)	Points	% of Grade
Exam 1, Oct 12, 2026	100	50%
Exam 2, Nov 30, 2026	100	50%
TOTAL		100%

Grading Scale

The final letter grade is based on a curve.

Course-specific Policies (Assignment Submission, Grading Timeline, Late work, and Technology)

All deadlines are firm. No late submissions or e-mail submissions are accepted for class participation.

Attendance

Based on prior experience, students who do not attend lectures perform poorly on exams. CSCI 585 accommodates student athletes with approved Travel Request Letters and students who give advance notice of religious observation.

Assigned Reading

Week 1: Introductions, ER Data Model & Relational Data Model

Assigned reading: Chen, Peter (March 1976). "The Entity-Relationship Model - Toward a Unified View of Data". ACM Transactions on Database Systems. **1** (1): 9-36.
<https://dl.acm.org/doi/pdf/10.1145/320434.320440>

Discussion

W. Kent. 1983. A Simple Guide to Five Normal Forms in Relational Database Theory. Communications of the ACM, Volume 25, Issue 2, February 1983.
<https://doi.org/10.1145/358024.358054>

D. D. Chamberlin, "Early History of SQL," in IEEE Annals of the History of Computing, vol. 34, no. 4, pp. 78-82, Oct.-Dec. 2012, doi:
<https://doi.org/10.1109/MAHC.2012.61>

Week 2: Labor Day

Week 3: Parallel Relational Database Management Systems

Assigned Reading: D. J. DeWitt, S. Ghandeharizadeh, D. A. Schneider, A. Bricker, H. -I. Hsiao and R. Rasmussen, "The Gamma database machine project," in IEEE Transactions on Knowledge and Data Engineering, vol. 2, no. 1, pp. 44-62, March 1990, doi:
<https://doi.org/10.1109/69.50905>

Discussion

Transactions and Crash Recovery (Presented by CSCI 585 TA Quankai Gao). Textbook material, e.g., Look at Chapters 15 and 17 of *Database Systems Concepts*, 4th edition, by Silberschatz, Korth, Sudarshan, publisher Morgan Kaufmann.

Week 4: Transactional Storage Managers

Assigned Reading: Jingyu Zhou, Meng Xu, Alexander Shraer, Bala Namasivayam, Alex Miller, Evan Tschannen, Steve Atherton, Andrew J. Beamon, Rusty Sears, John Leach, Dave Rosenthal, Xin Dong, Will Wilson, Ben Collins, David Scherer, Alec Grieser, Young Liu, Alvin Moore, Bhaskar Muppana, Xiaoge Su, and Vishesh Yadav. 2022. FoundationDB: A Distributed Key Value Store. SIGMOD Rec. 51, 1 (March 2022), 24-31. <https://doi.org/10.1145/3542700.3542707>

Discussion

Ghandeharizadeh, S., Bernstein, P.A., Borthakur, D., Huang, H., Menon, J., Puri, S. (2023). Disaggregated Database Management Systems. In: Nambiar, R., Poess, M. (eds) Performance Evaluation and Benchmarking. TPCTC 2022. Lecture Notes in Computer Science, vol 13860. Springer, Cham. From the USC secure network, use the following link to access the paper:
https://libcatalog.usc.edu/discovery/fulldisplay?context=PC&vid=01USC_INST:01USC&search_scope=MyInst_and_CI&tab=Everything&docid=cdi_scopus_primary_2_s2_0_85152533804

Shahram Ghandeharizadeh and Hieu Nguyen. 2019. Design, Implementation, and Evaluation of Write-Back Policy with Cache Augmented Data Stores. VLDB 2019.
<https://www.vldb.org/pvldb/vol12/p836-ghandeharizadeh.pdf>

Week 5: Physical Data Design

Assigned Reading: Shahram Ghandeharizadeh and David J. DeWitt. 1990. Hybrid-Range Partitioning Strategy: A New Declustering Strategy for Multiprocessor Database Machines. In Proceedings of the 16th International Conference on Very Large Data Bases (VLDB '90). Morgan Kaufmann Publishers Inc., San Francisco, CA, USA, 481-492. <https://www.vldb.org/conf/1990/P481.PDF>

Discussion

Transactions: Concurrency Control Protocols

Week 6: Benchmarks

Assigned Reading: S. Barahmand and S. Ghandeharizadeh. BG: A Benchmark to Evaluate Interactive Social Networking Actions. Proceedings of the biennial Conference on Innovative Data Systems Research (CIDR), Asilomar, California, January 6-9, 2013. https://www.cidrdb.org/cidr2013/Papers/CIDR13_Paper93.pdf

Discussion

Brian F. Cooper, Adam Silberstein, Erwin Tam, Raghu Ramakrishnan, and Russell Sears. 2010. Benchmarking cloud serving systems with YCSB. In Proceedings of the 1st ACM symposium on Cloud computing (SoCC '10). Association for Computing Machinery, New York, NY, USA, 143-154. <https://doi.org/10.1145/1807128.1807152>

Week 7: Graph Query Processing

Assigned Reading: Chen, Hongzhi and Li, Changji and Fang, Juncheng and Huang, Chenghuan and Cheng, James and Zhang, Jian and Hou, Yifan and Yan, Xiao. Grasper: A High Performance Distributed System for OLAP on Property Graphs, 2019, Proceedings of the ACM Symposium on Cloud Computing. <https://doi.org/10.1145/3357223.3362715>

Week 8: Exam 1

Exam 1 covers Week 1-7 assigned readings.

Week 9: Separation of Storage from Processing

Assigned Reading: Jasraj Dange, Andrei Dragus, Ali Ghodsi, Haoyu Huang, Yihe Huang, Stas Kelvich, Heikki Linnakangas, Hans Norheim, Ippokratis Pandis, Nikita Shamgunov, Em Sharnoff, John Spray, Zhou Sun, Reynold Xin, Matei Zaharia. Lakebase: Serverless Postgres over Open Lake Storage. VLDB 2026

Discussion

Haoyu Huang and Shahram Ghandeharizadeh. 2021. Nova-LSM: A Distributed, Component-based LSM-tree Key-value Store. In

Proceedings of the 2021 International Conference on Management of Data (SIGMOD '21). Association for Computing Machinery, New York, NY, USA, 749–763. <https://doi.org/10.1145/3448016.3457297>

Week 10: Memory Management

Assigned Reading: Shahram Ghandeharizadeh, Sandy Irani, Jenny Lam, and Jason Yap. 2014. CAMP: A Cost Adaptive Multi-Queue Eviction Policy for Key-Value Stores. In Proceedings of the 15th International Middleware Conference (Middleware '14). Association for Computing Machinery, New York, NY, USA, 289–300.

<https://doi.org/10.1145/2663165.2663317>

Discussion

Elizabeth J. O'Neil, Patrick E. O'Neil, and Gerhard Weikum. 1993. The LRU-K page replacement algorithm for database disk buffering. SIGMOD Rec. 22, 2 (June 1, 1993), 297–306. <https://doi.org/10.1145/170036.170081>

Week 11: Raft & PAXOS

Assigned Reading: Diego Ongaro and John Ousterhout. 2014. In search of an understandable consensus algorithm. In Proceedings of the 2014 USENIX conference on USENIX Annual Technical Conference (USENIX ATC'14). USENIX Association, USA, 305–320.

Optional reading:

Zhiying Liang, Vahab Jabrayilov, Abutalib Aghayev, Aleksey Charapko. 2025. HoliPaxos: Towards More Predictable Performance in State Machine Replication. VLDB, Volume 18, No. 8, 2025. <https://www.vldb.org/pvldb/vol18/p2505-charapko.pdf>
<https://doi.org/10.14778/3742728.3742744>

Week 12: Cache Augmented Database Management Systems

Assigned Reading: Shahram Ghandeharizadeh, Jason Yap, and Hieu Nguyen. 2014. Strong consistency in cache augmented SQL systems. Proceedings of the 15th International Middleware Conference (Middleware '14). Association for Computing Machinery, New York, NY, USA, 181–192.

<https://doi.org/10.1145/2663165.2663318> OR using USC DL
<https://dl-acm-org.libproxy1.usc.edu/doi/pdf/10.1145/2663165.2663318>

Discussion

Shahram Ghandeharizadeh, Sandy Irani, and Jenny Lam. 2015. Cache replacement with memory allocation. In Proceedings of the Meeting on Algorithm Engineering & Experiments (ALENEX '15). Society for Industrial and Applied Mathematics, USA, 1–9. <https://epubs.siam.org/doi/10.1137/1.9781611973754.1>

Week 13: MapReduce

Assigned Reading: Jeffrey Dean and Sanjay Ghemawat.
MapReduce: Simplified Data Processing on Large Clusters.
Sixth Symposium on Operating System Design and
Implementation (OSDI) 2004.
<https://dl.acm.org/doi/pdf/10.1145/1327452.1327492>

Discussion

Juwei Shi, Yunjie Qiu, Umar Farooq Minhas, Limei Jiao, Chen Wang, Berthold Reinwald, and Fatma Özcan. 2015. Clash of the titans: MapReduce vs. Spark for large scale data analytics. Proc. VLDB Endow. 8, 13 (September 2015), 2110–2121.
<https://doi.org/10.14778/2831360.2831365>

Week 14: RAID

Assigned Reading: David A. Patterson, Garth Gibson, and Randy H. Katz. 1988. A case for redundant arrays of inexpensive disks (RAID). In Proceedings of the 1988 ACM SIGMOD international conference on Management of data (SIGMOD '88). Association for Computing Machinery, New York, NY, USA, 109–116. <https://doi.org/10.1145/50202.50214>

Week 15: Course Evaluation, Review, and Exam 2

Exam 2 covers assigned readings of Week 9–14.

Fall 2026 Course Schedule: A Weekly Breakdown

	Topics/Daily Activities	Readings/Preparation	Key Concepts
Week 1 8/24	Introduction, SDM, Relational	ER, Conceptual model	Conceptual data models
Week 2 8/31	Parallel DBMSs	Gamma	Parallel query processing, crash recovery, shared-nothing architecture
Week 3 9/7	Labor Day	No Class	No Class
Week 4 9/14	Xact Storage Managers	FoundationDB Guest lecture by Dr. Hieu Nguyen of eBay	OCC, MVCC, Fat Clients, DBMS as a collection of microservices
Week 5 9/21	Data Declustering	Hybrid Range Partitioning Strategy	Sharding, Load Balancing
Week 6 9/28	Benchmarks	BG Social Networking Benchmark	Performance metrics, Closed and Open Simulation Models

Week 7 10/5	Graph Query Processing	Grasper	Graph data model, Gremlin query language, operators
Week 8 10/12	Review for Exam 1 (1 hour) + 1 hour Exam 1 is on the same day	Exam 1	Review for Exam 1 + Exam 1.
Week 9 10/19	Disaggregated DBMSs	Lakebase Guest lecture by Dr. Haoyu Huang of Databricks	Separation of storage from processing, shared disk architecture.
Week 10 10/26	Memory Mgmt	CAMP and LRU	Low complexity memory mgmt algorithms that consider size and cost.
Week 11 11/2	Raft & Paxos	Raft consensus algorithm	Distributed consensus and an algorithm to implement it.
Week 12 11/9	Cache Augmented Database Mgmt Systems	IQ Framework	Impact of caches on transactions
Week 13 11/16	MapReduce	Programming paradigm for parallelism.	Divide-and-Conquer, Resilient to failures.
Week 14 11/23	Redundant arrays of disks	RAID	Parity, availability, Amdahl's law, small & large I/O.
Week 15 11/30	Course Evals, Review for Exam 2 & + 1 hour Exam 2	Review + Exam 2	Course Evaluation, Review for Exam 2 + Exam 2

Time Permitting: NoSQL Database Systems

Assigned Reading: Joseph Idziorek and Alex Keyes and Colin Lazier and Somu Perianayagam and Prithvi Ramanathan and James Christopher Sorenson III and Doug Terry and Akshat Vig. 2023. Distributed Transactions at Scale in Amazon DynamoDB. 2023 USENIX Annual Technical Conference (USENIX ATC 23) .

<https://www.usenix.org/conference/atc23/presentation/idziorek>

Discussion

Mostafa Elhemali and Niall Gallagher and Nick Gordon and Joseph Idziorek and Richard Krog and Colin Lazier and Erben Mo and Akhilesh Mritunjai and Somasundaram Perianayagam and Tim Rath and Swami Sivasubramanian and James Christopher Sorenson III and Sroaj Sosothikul and Doug Terry and Akshat Vig. 2022. Amazon DynamoDB: A Scalable, Predictably Performant, and Fully Managed NoSQL Database Service, 2022 USENIX Annual Technical Conference (USENIX ATC 22) .

<https://www.usenix.org/conference/atc22/presentation/elhemali>

Jingyu Zhou, Meng Xu, Alexander Shraer, Bala Namasivayam, Alex Miller, Evan Tschannen, Steve Atherton, Andrew J. Beamon, Rusty Sears, John Leach, Dave

Rosenthal, Xin Dong, Will Wilson, Ben Collins, David Scherer, Alec Grieser, Young Liu, Alvin Moore, Bhaskar Muppana, Xiaoge Su, Vishesh Yadav:
FoundationDB: A Distributed Unbundled Transactional Key Value Store. SIGMOD Conference 2021: 2653-2666 <https://www.foundationdb.org/files/fdb-paper.pdf>

Statement on Academic Conduct and Support Systems

Academic Conduct:

Plagiarism – presenting someone else’s ideas as your own, either verbatim or recast in your own words – is a serious academic offense with serious consequences. Please familiarize yourself with the discussion of plagiarism in SCampus in Part B, Section 11, “Behavior Violating University Standards” policy.usc.edu/scampus-part-b. Other forms of academic dishonesty are equally unacceptable. See additional information in SCampus and university policies on Research and Scholarship Misconduct.

Students and Disability Accommodations:

USC welcomes students with disabilities into all of the University’s educational programs. The Office of Student Accessibility Services (OSAS) is responsible for the determination of appropriate accommodations for students who encounter disability-related barriers. Once a student has completed the OSAS process (registration, initial appointment, and submitted documentation) and accommodations are determined to be reasonable and appropriate, a Letter of Accommodation (LOA) will be available to generate for each course. The LOA must be given to each course instructor by the student and followed up with a discussion. This should be done as early in the semester as possible as accommodations are not retroactive. More information can be found at osas.usc.edu. You may contact OSAS at (213) 740-0776 or via email at osasfrontdesk@usc.edu.

Support Systems:

Counseling and Mental Health - (213) 740-9355 – 24/7 on call
studenthealth.usc.edu/counseling

Free and confidential mental health treatment for students, including short-term psychotherapy, group counseling, stress fitness workshops, and crisis intervention.

National Suicide Prevention Lifeline - 1 (800) 273-8255 – 24/7 on call
suicidepreventionlifeline.org

Free and confidential emotional support to people in suicidal crisis or emotional distress 24 hours a day, 7 days a week.

Relationship and Sexual Violence Prevention Services (RSVP) - (213) 740-9355(WELL), press “0” after hours – 24/7 on call

studenthealth.usc.edu/sexual-assault

Free and confidential therapy services, workshops, and training for situations related to gender-based harm.

Office for Equity, Equal Opportunity, and Title IX (EEO-TIX) - (213) 740-5086
eeotix.usc.edu

Information about how to get help or help someone affected by harassment or discrimination, rights of protected classes, reporting options, and additional resources for students, faculty, staff, visitors, and applicants.

Reporting Incidents of Bias or Harassment - (213) 740-5086 or (213) 821-8298

usc-advocate.symplicity.com/care_report

Avenue to report incidents of bias, hate crimes, and microaggressions to the Office for Equity, Equal Opportunity, and Title for appropriate investigation, supportive measures, and response.

The Office of Student Accessibility Services (OSAS) - (213) 740-0776

osas.usc.edu

OSAS ensures equal access for students with disabilities through providing academic accommodations and auxiliary aids in accordance with federal laws and university policy.

USC Campus Support and Intervention - (213) 821-4710

campussupport.usc.edu

Assists students and families in resolving complex personal, financial, and academic issues adversely affecting their success as a student.

Diversity, Equity and Inclusion - (213) 740-2101

diversity.usc.edu

Information on events, programs and training, the Provost's Diversity and Inclusion Council, Diversity Liaisons for each academic school, chronology, participation, and various resources for students.

USC Emergency - UPC: (213) 740-4321, HSC: (323) 442-1000 – 24/7 on call

dps.usc.edu, emergency.usc.edu

Emergency assistance and avenue to report a crime. Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible.

USC Department of Public Safety - UPC: (213) 740-6000, HSC: (323) 442-120 – 24/7 on call

dps.usc.edu

Non-emergency assistance or information.

Office of the Ombuds - (213) 821-9556 (UPC) / (323) 442-0382 (HSC)

ombuds.usc.edu

A safe and confidential place to share your USC-related issues with a University Ombuds who will work with you to explore options or paths to manage your concern.

Occupational Therapy Faculty Practice - (323) 442-3340 or otfp@med.usc.edu

chan.usc.edu/otfp

Confidential Lifestyle Redesign services for USC students to support health promoting habits and routines that enhance quality of life and academic performance.